Partial Identification

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Motivation I

The Limits of Causal Inference

- (Y_0, Y_1) never observed for the same person $\Rightarrow$ cannot learn their joint distribution.
- Quantities like $\mathrm{Var}(Y_1 - Y_0)$ or $P(Y_1 - Y_0 > 0)$ are not identifiable.
- Even to learn the ATE, we have to make untestable assumptions.

Partial Identification

- Even if we cannot pin down a parameter of interest θ exactly, we may be able to rule out many values.
- Rather than an identification strategy, it summarises several approaches to learn something about the data under **weaker assumptions**.
- The best way to get a feeling of it is to go through several examples.

Motivation II

Outline

1. ** A simple example: Reverse regression bounds
2. Bounds on ATE allowing for selection bias (Blundell, Gosling, Ichimura & Meghir, 2007).
3. Bounds for Sample selection (Lee, 2009)
4. ** Bounding the distribution of treatment effects (Fan & Park, 2010).
5. Using the MTE to bound additional parameters (Mogstad, Santos & Torgovitsky, 2018).

** *Based on the slides of Frank Ditraglia.*

Simple Example: Reverse Regression Bounds

Population linear regression

Let α and β be the intercept and slope from the population linear regression of Y on X. Thus we can write

$$Y = \alpha + \beta X + U,$$

where

$$\beta \equiv \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \qquad \alpha \equiv E[Y] - \beta E[X], \qquad U \equiv Y - \alpha - \beta X.$$

By construction,

$$E(XU) = 0, \qquad E(U) = 0.$$

Point identification

- If we could observe the whole population from which our sample was drawn, could we uniquely determine the parameters of interest?
- Suppose we observe the joint distribution of (X, Y).
- This is enough information to calculate (α, β) explicitly: they are point identified.

Classical Measurement Error

- Suppose we observe $(Y, \tilde{X})$ rather than (Y, X), where

$$\tilde{X} = X + W.$$

- W is classical measurement error:

$$\text{Cov}(W, X) = 0, \quad \text{Cov}(W, U) = 0, \quad E(W) = 0.$$

- Are α and β still point identified?

The Good News: $E(\tilde{X}) = E(X + W) = E(X)$.

$$\begin{aligned}
\text{Cov}(\tilde{X}, Y) &= \text{Cov}(X + W, Y) \\
&= \text{Cov}(X, Y) + \text{Cov}(W, Y) \\
&= \text{Cov}(X, Y) + \text{Cov}(W, \alpha + \beta X + U) \\
&= \text{Cov}(X, Y) + \text{Cov}(W, U) + \beta\,\text{Cov}(W, X) \\
&= \text{Cov}(X, Y).
\end{aligned}$$

Are α and β Still Point Identified?

The Bad News

- Because $\text{Var}(W)$ is not point identified, neither are α and β.

$$\text{Var}(\tilde{X}) = \text{Var}(X + W) = \text{Var}(X) + \text{Var}(W) \ \geq \ \text{Var}(X).$$

$$\beta \equiv \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\text{Cov}(\tilde{X}, Y)}{\text{Var}(\tilde{X}) - \text{Var}(W)}, \qquad \alpha \equiv E[Y] - \beta E[X] = E[Y] - \beta E[\tilde{X}].$$

Partial Identification

- We can still bound β and hence α: the so-called *reverse regression bounds*.

A Lower Bound for β

- Since $\mathrm{Cov}(X, Y) = \mathrm{Cov}(\tilde{X}, Y)$,

$$\frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X) + \mathrm{Var}(W)} = \frac{\mathrm{Cov}(X, Y)/\mathrm{Var}(X)}{1 + \mathrm{Var}(W)/\mathrm{Var}(X)} = \frac{\beta}{1 + \mathrm{Var}(W)/\mathrm{Var}(X)}.$$

- Since $\mathrm{Var}(W)/\mathrm{Var}(X)$ is non-negative,

$$\left| \frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})} \right| \leq |\beta|.$$

An Upper Bound for β

- Run the reverse regression $\tilde{X}$ on Y:

$$\frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(Y)} = \frac{\mathrm{Cov}(X, Y)}{\beta^2 \,\mathrm{Var}(X) + \mathrm{Var}(U)} = \frac{\beta \,\mathrm{Var}(X)}{\beta^2 \,\mathrm{Var}(X) + \mathrm{Var}(U)}.$$

- Take the reciprocal:

$$\frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} = \beta + \frac{\mathrm{Var}(U)}{\beta \,\mathrm{Var}(X)} = \beta \left[1 + \frac{\mathrm{Var}(U)}{\beta^2 \,\mathrm{Var}(X)} \right].$$

- The bracketed factor exceeds one, so

$$\left| \frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} \right| \geq |\beta|.$$

Reverse Regression Bounds

Terminology

- A bound is *sharp* if it cannot be improved under our assumptions.
- A bound is *tight* if it is short enough to be useful in a practical example.

Assumptions

- $Y = \alpha + \beta X + U$ where $E(XU) = E(U) = 0$.
- Observe $(\tilde{X}, Y)$.
- $\tilde{X} = X + W$ with $E(W) = 0$, $\mathrm{Cov}(W, X) = 0$, $\mathrm{Cov}(W, U) = 0$.

Sharp Bounds for β

$$\beta \in \left[\frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})}, \frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} \right].$$

How Tight Are the Reverse Regression Bounds?

- Let r denote the correlation between $\tilde{X}$ and Y. Then:

$$r^2 \equiv \frac{\mathrm{Cov}(\tilde{X}, Y)^2}{\mathrm{Var}(\tilde{X})\,\mathrm{Var}(Y)} = \frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})} \cdot \frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(Y)}.$$

- Re-arranging, it follows that:

$$r^2 \cdot \frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} = \frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})}.$$

- All else equal, bounds for β are tighter when $\tilde{X}$ and Y are strongly correlated:

$$\text{Width} = \left| \frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} - \frac{\mathrm{Cov}(\tilde{X}, Y)}{\mathrm{Var}(\tilde{X})} \right| = (1 - r^2) \left| \frac{\mathrm{Var}(Y)}{\mathrm{Cov}(\tilde{X}, Y)} \right|.$$

Simulation Setup

```r
library(tidyverse)
library(broom) # for tidy()

set.seed(1066)
n <- 5000
X <- rnorm(n)
U <- rnorm(n)
W <- rnorm(n)

alpha <- 0.5
beta  <- 1

Y <- alpha + beta * X + U
Xtilde <- X + W
```

```r
c(
  forward = cov(Xtilde, Y) / var(Xtilde),
  truth   = beta,
  reverse = var(Y) / cov(Xtilde, Y)
) |> round(2)
```

```
## forward   truth reverse
##    0.51    1.00    1.95
```

```r
# True Regression (Not Feasible in Practice)
lm(Y ~ X) |> tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)    0.489    0.0140      34.8 9.56e-238
## 2 X              1.02     0.0138      73.9 0
```

```r
# Reduce correlation between X and Y
Y2 <- alpha + beta * X + 3 * U

c(  forward = cov(Xtilde, Y2) / var(Xtilde),
  truth   = beta,
  reverse = var(Y2) / cov(Xtilde, Y2)
) |> round(2)
```

```
## forward   truth reverse
##    0.52    1.00    9.31
```

```r
lm(Y2 ~ X) |> tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)    0.466    0.0421      11.1 3.95e- 28
## 2 X              1.07     0.0414      25.7 7.45e-137
```

Blundell, Gosling, Ichimura & Meghir (2007) I

- These slides summarize Sections 1–2 of: Blundell, R., Gosling, A., Ichimura, H., & Meghir, C. (2007). *Changes in the distribution of male and female wages accounting for employment composition using bounds*. Econometrica, 75(2), 323-363.

- **Rising inequality (UK/US, last 30–40 years):** larger IQR of log hourly wages among workers; higher returns to education; cohort effects; greater returns to unobserved skill.
- **Employment composition shifts:** male employment (22–59) fell (93% $\rightarrow$ 83% from 1978 to 2000); female employment rose, especially among the more educated and younger women.
- **Institutions & policy:** early 1980s demand shocks; welfare and disability programs; housing benefit & rent reforms $\Rightarrow$ higher replacement rates, affecting participation and composition.

Blundell, Gosling, Ichimura & Meghir (2007) II

- **Selection problem:** observed wage distribution is *censored* by employment decisions; need methods robust to nonrandom selection.
- **Paper's aim:** develop and apply *bounds* for the wage distribution and inequality allowing for selection, tightening worst-case bounds via theory-motivated restrictions.

Figure 1: Interquartile Range of Log Wages

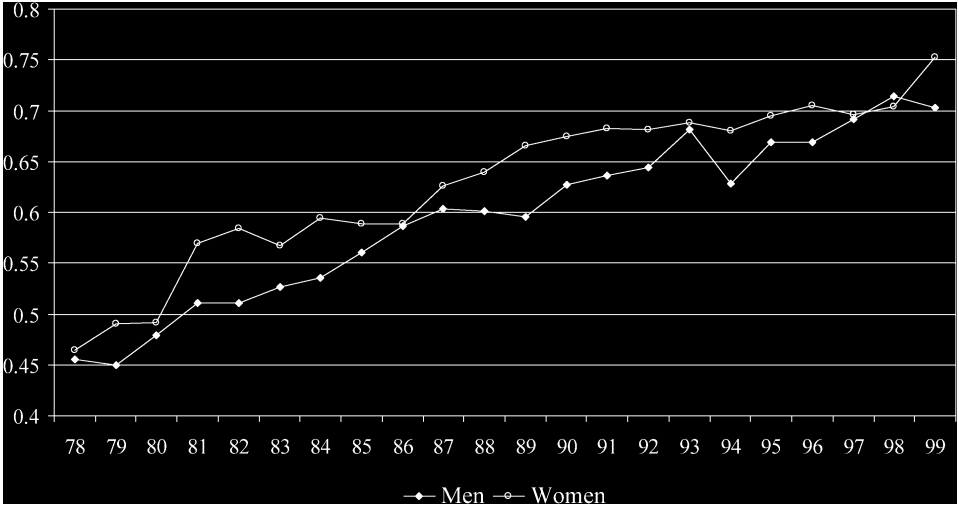
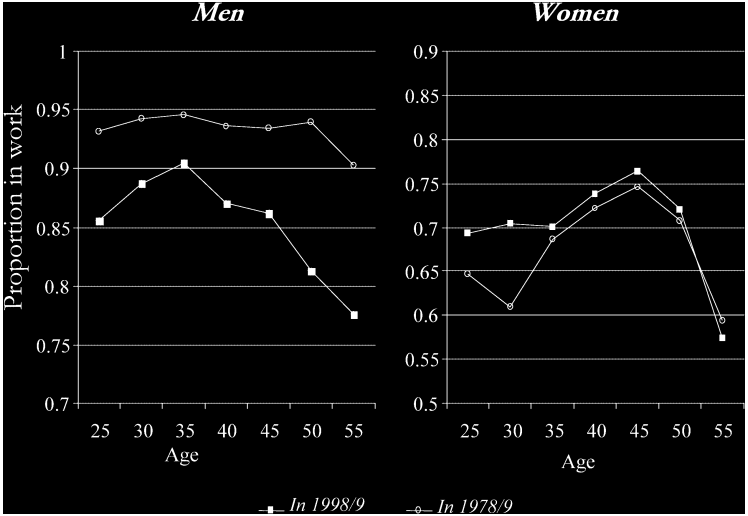


Figure 2: Employment by Age in 1978 and 1998



Setup

Let W be (log) wage, X characteristics (gender, age, education, year), and $E \in \{0, 1\}$ an employment indicator.

The target CDF (unobserved distribution of wages in the population with $X = x$):

$$F(w \mid x) \;=\; F(w \mid x, E{=}1)\, P(x) \;+\; F(w \mid x, E{=}0)\, [1 - P(x)],$$
$$\text{where } P(x) \equiv \Pr(E{=}1 \mid X{=}x).$$

Only $F(w \mid x, E{=}1)$ and $P(x)$ are identified in the data; $F(w \mid x, E{=}0)$ is not.

Worst-Case (Manski) Bounds

Using only $0 \leq F(w \mid x, E{=}0) \leq 1$, we obtain an identification region:

$$\underline{F}(w \mid x) := F(w \mid x, E{=}1)P(x),$$
$$\overline{F}(w \mid x) := F(w \mid x, E{=}1)P(x) + [1 - P(x)],$$

so that

$$\underline{F}(w \mid x) \;\leq\; F(w \mid x) \;\leq\; \overline{F}(w \mid x).$$

Implication for quantiles $w_q(x)$: solve

$$q = F(w \mid x, E{=}1)P(x) \quad \text{(upper bound for } w_q\text{)},$$
$$q = F(w \mid x, E{=}1)P(x) + [1 - P(x)] \quad \text{(lower bound for } w_q\text{)}.$$

Restriction I: Stochastic Dominance (Positive Selection)

Assume workers' wage distribution first-order stochastically dominates nonworkers':

$$F(w \mid x, E{=}1) \;\leq\; F(w \mid x, E{=}0) \quad \forall\, w, x.$$

Equivalent form (via Bayes' rule):

$$\Pr(E{=}1 \mid W \leq w, X{=}x) \;\leq\; \Pr(E{=}1 \mid W > w, X{=}x).$$

Then the bounds tighten to

$$F(w \mid x, E{=}1) \;\leq\; F(w \mid x) \;\leq\; F(w \mid x, E{=}1)P(x) + [1 - P(x)].$$

Restriction II: Median Restriction (Weaker Positive Selection)

Require the median of workers' wages to be at least that of nonworkers:

$$w_{50}(E{=}1 \mid x) \;\geq\; w_{50}(E{=}0 \mid x).$$

Equivalent bounds on $F(w \mid x, E{=}0)$:

$$\begin{cases} 0 \leq F(w \mid x, E{=}0) \leq 1, & \text{if } w < w_{50}(E{=}1 \mid x), \\ 0.5 \leq F(w \mid x, E{=}0) \leq 1, & \text{if } w \geq w_{50}(E{=}1 \mid x). \end{cases}$$

Insert these into the mixture equation to tighten bounds, especially at/above the median.

Restriction III: Exclusion (Manski, 1994)

Let Z affect employment but not wages: $F(w \mid x, z) = F(w \mid x)$. Then

$$\max_z \{F(w \mid x, z, E{=}1)P(x, z)\} \;\leq\; F(w \mid x) \;\leq\; \min_z \{F(w \mid x, z, E{=}1)P(x, z) + 1 - P(x, z)\}.$$

Intuition:
Pick z values that maximize $\Pr(E{=}1 \mid W \leq w, X{=}x, Z{=}z)$ for the lower bound and $\Pr(E{=}1 \mid W > w, X{=}x, Z{=}z)$ for the upper bound.

Restriction IV: Monotonicity (Manski–Pepper, 2000)

Weaker than exclusion:
Assume $F(w \mid x, z') \leq F(w \mid x, z)$ whenever $z < z'$ (first-order stochastic dominance in z). Define

$$F^{\ell}(w \mid x, z_1) := \max_{z \geq z_1} \{F(w \mid x, z, E{=}1)P(x, z)\},$$
$$F^{u}(w \mid x, z_1) := \min_{z \leq z_1} \{F(w \mid x, z, E{=}1)P(x, z) + 1 - P(x, z)\}.$$

Integrate over the empirical distribution of $Z \mid X{=}x$ to obtain

$$\mathbb{E}_Z\Big[F^{\ell}(w \mid x, Z) \mid x\Big] \;\leq\; F(w \mid x) \;\leq\; \mathbb{E}_Z[F^{u}(w \mid x, Z) \mid x]\,.$$

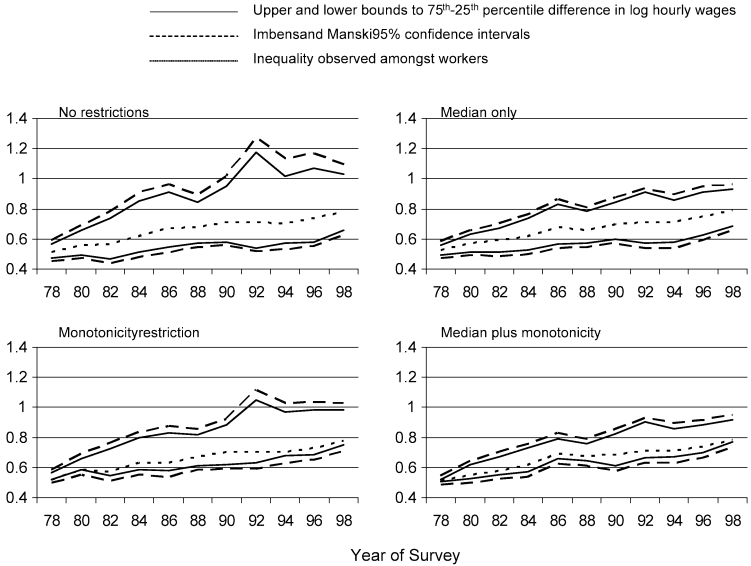
Bounds on Within-Group Inequality (e.g., IQR)

Let $D(x) = w_{q_2}(x) - w_{q_1}(x)$ (e.g., q_1=0.25, q_2=0.75). With the CDF constraints implied by the mixture,

$$
\begin{aligned}
D_{\mathrm{LB}}(x) &= \max\{0,\ w_{q_2}^{(l)}(x) - w_{q_1}^{(u)}(x)\}, \\
D_{\mathrm{UB}}(x) &= \sup_{w_0 \in [w_{q_1}^{(l)},\, w_{q_1}^{(u)}]} \left[F^{-1}\Big(F(w_0 \mid x, E{=}1) + \tfrac{q_2 - q_1}{P(x)} \;\Big|\; x, E{=}1 \Big) - w_0 \right].
\end{aligned}
$$

These use the "slowest feasible increase" in $F(\cdot \mid x)$ parallel to $P(x)F(\cdot \mid x, E{=}1)$.

Bounds under several assumptions



Lee bounds (partial identification for sample selection)

Lee, David S. "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." The Review of Economic Studies 76, no. 3 (2009): 1071-1102.

Lee bounds are a **partial-identification** result for treatment effects when you have **sample selection / attrition** that differs by treatment status.

They give **sharp bounds** on the treatment effect under a credible monotonicity restriction, instead of forcing point identification with strong assumptions.

Setup: treatment, outcome, and selection

You run an experiment (or have quasi-experimental treatment) with:

- $D \in \{0, 1\}$: treatment assignment (often randomised).
- Y: outcome of interest (e.g., earnings).
- $S \in \{0, 1\}$: indicator that Y is observed (e.g., stays in the sample / responds).

The problem: even if D is randomised, we only observe Y when $S = 1$. If treatment affects selection (S), then the naive comparison

$$E[Y \mid D = 1, S = 1] - E[Y \mid D = 0, S = 1]$$

is generally biased for any causal estimand of interest.

Key assumption: monotonic selection (no "defiers")

Lee (2009) imposes a monotonicity condition on selection:

$$S_1 \geq S_0 \quad \text{a.s.}$$

where S_d denotes the potential selection indicator under treatment $d \in \{0, 1\}$.

Interpretation:

- Treatment can only *weakly increase* the chance that outcomes are observed (or, in the opposite version, only weakly decrease it).
- There are no units who would be observed under control but unobserved under treatment (no "selection defiers").

This is analogous in spirit to monotonicity used for LATE, but applied to *selection/attrition*.

Who is identified? The "always-observed" group

Define the principal stratum:

- **Always-observed**: $S_1 = 1$ and $S_0 = 1$.

Lee bounds identify (i.e. bound) the treatment effect for this group:

$$\text{ATE}_{AO} \;=\; E[Y_1 - Y_0 \mid S_1 = 1,\, S_0 = 1].$$

Why this group?
Because it is the only group for which, conceptually, both potential outcomes are observable under both treatment states (so one can meaningfully talk about the causal effect for that group).

Main idea: trimming to re-balance selection rates I

Let the observed selection rates be

$$p_1 = P(S = 1 \mid D = 1), \qquad p_0 = P(S = 1 \mid D = 0).$$

Assume $p_1 \geq p_0$ (consistent with $S_1 \geq S_0$). Then the treated group has *more observed outcomes*. Under monotonicity, the **extra observed treated** units must be those with $(S_1 = 1, S_0 = 0)$ — they are observed only if treated, and **should not be in the always-observed** comparison.

Main idea: trimming to re-balance selection rates II

Lee's idea:

1. Compute the fraction to retain:

$$\alpha = \frac{p_0}{p_1} \in (0, 1].$$

2. Trim the treated outcome distribution so that only an α fraction of the treated observations remain. The trimming can be done in two opposite (worst/best) ways to get bounds.

Lower and upper bounds (case $p_1 \geq p_0$)

Let $Y^{(1)}$ denote observed outcomes in the treated sample $(D = 1, S = 1)$ and write $q_1(\tau)$ for the τ-quantile of that distribution.

Lower bound (LB). Trim away the *top* $(1 - \alpha)$ of treated outcomes (i.e. assume the extra treated-only observed units are the *best* outcomes). Then

$$\text{LB} \;=\; E\big[Y \mid D = 1, S = 1,\; Y \leq q_1(\alpha)\big] \;-\; E\big[Y \mid D = 0, S = 1\big].$$

Upper bound (UB). Trim away the *bottom* $(1 - \alpha)$ of treated outcomes (i.e. assume the extra treated-only observed units are the *worst* outcomes). Then

$$\text{UB} \;=\; E\big[Y \mid D = 1, S = 1,\; Y \geq q_1(1 - \alpha)\big] \;-\; E\big[Y \mid D = 0, S = 1\big].$$

Intuition: because we do not know whether the additional observed treated units are "good" or "bad", we bound the effect by assigning them in the most favorable vs. least favorable consistent way.

Converse case: $p_1 < p_0$

If instead $p_1 < p_0$ and you believe $S_1 \leq S_0$, perform the analogous trimming on the *control* group:

- compute $\alpha = p_1/p_0$,
- trim the control outcome distribution to an α fraction,
- form symmetric bounds by comparing trimmed controls to observed treated outcomes.

(Everything above holds with roles of treatment/control exchanged.)

Sharpness

- The interval $[\mathrm{LB}, \mathrm{UB}]$ is *valid* under random assignment of D plus the monotone selection assumption.
- It is also *sharp*: without further assumptions you generally cannot shrink the interval because there exist data-generating processes satisfying the assumptions that attain the endpoints.

Practical notes

- **Conditional trimming:** Often implemented conditional on covariates X. Compute $p_1(x), p_0(x)$ within cells (or use estimated propensity scores), trim within cells, then average — this tightens bounds.
- **Inference:** Trimming uses estimated quantiles and non-smooth operators. Applied work typically uses the bootstrap or asymptotic theory tailored for partially identified parameters (careful bootstrap; subsampling; methods for moment inequalities).
- **Implementation detail:** When trimming, decide how to break ties at the quantile and document the choice; with discrete Y you may need randomized trimming or conservative adjustments.

When Lee bounds are useful & limitations

Examples of use

- Randomized trials with *differential attrition* by assignment.
- Programs that change *participation/employment* so outcomes are observed only for participants.
- Survey nonresponse that is affected by treatment.

Limitations

- Monotonicity on selection is substantive: if treatment plausibly makes some individuals less likely and others more likely to be observed, the assumption fails.
- Bounds can be wide when selection rates differ a lot or when outcome variance is large.

Simulation set up

Please simulate the following process:

$$
\begin{aligned}
Y_0 &= \mu_0 + U_0, \text{ where } U_0 \sim N(0,1) \\
Y_1 &= Y_0 + \mu_1 + U_1 \text{ where } U_1 \sim N(0,1) \\
D &= \text{Bernouli}(0.5) \\
S_0 &= \text{rank}(Y_0) \leq p_0 \text{ with } p_0 = 0.6 \\
S_1 &= \text{rank}(Y_0) \leq p_1 \text{ with } p_1 = 0.8 \\
D &= S_0(1-D) + S_1 D \\
Y &= Y_0(1-D) + Y_1 D \text{ if } S = 1
\end{aligned}
$$

Set ATE $= mu_1 = 0.3$. check the naive ATE (ignoring attrition).

Simulation setup

```r
# ---------- Parameters ----------
set.seed(9237)
n <- 5000
p_treat <- 0.5
mu0 <- 0
mu1 <- 0.3
sigma <- 1

p0_base <- 0.6
p1_base <- 0.8
D <- rbinom(n, 1, p_treat)
Y0 <- rnorm(n, mean = mu0, sd = sigma)
Y1 <- Y0 + mu1 + rnorm(n, mean = 0, sd = 0.2)
```

We construct a common latent rank based on Y_0: units with highest Y_0 are more likely to be missing. Then define thresholds so that $S_1 \geq S_0$ for every unit.

```r
# Create a rank-based latent score from Y0
rank_score <- rank(Y0) / (n + 1)

S0 <- as.integer(rank_score <= p0_base)
S1 <- as.integer(rank_score <= p1_base)

# Realised selection and observed Y
S <- ifelse(D==1, S1, S0)
Y_obs <- ifelse(S==1, ifelse(D==1, Y1, Y0), NA)
```

```r
cat("observed selection rates:",
    "p0 =", round(mean(S0),3),
    "p1 =", round(mean(S1),3), "\n")
```

```
## observed selection rates: p0 = 0.6 p1 = 0.8
```

```r
cat("Monotonicity holds for all units (S1 >= S0):", all(S1 >= S0), "\n")
```

```
## Monotonicity holds for all units (S1 >= S0): TRUE
```

```r
dat <- data.frame(D, Y0, Y1, S0, S1, S, Y = Y_obs)
head(dat)
```

```
##   D          Y0           Y1 S0 S1 S          Y
## 1 1  0.3313719  0.85634356  0  1 1  0.8563436
## 2 0 -2.1973681 -1.95164115  1  1 1 -2.1973681
## 3 1 -0.7163072 -0.40166603  1  1 1 -0.4016660
## 4 0 -1.2481017 -1.01286801  1  1 1 -1.2481017
## 5 0 -0.5631988 -0.07049906  1  1 1 -0.5631988
## 6 0 -0.5348711 -0.36228629  1  1 1 -0.5348711
```

Check naive estimator

```
Y_treat_obs <- dat$Y[dat$D==1 & dat$S==1]
Y_ctrl_obs  <- dat$Y[dat$D==0 & dat$S==1]

naive_diff <- mean(Y_treat_obs) - mean(Y_ctrl_obs)
cat("Naive observed difference:", round(naive_diff,3), "\n")
```

```
## Naive observed difference: 0.641
```

Compute Lee bounds

(1) Compute the trimming factor $\alpha = p_0/p_1$

(2) **Lower bound (LB).** Trim away the *top* $(1-\alpha)$ of treated outcomes (i.e. assume the extra treated-only observed units are the *best* outcomes). Then

$$\text{LB} \;=\; E\big[Y \mid D=1, S=1,\; Y \le q_1(\alpha)\big] \;-\; E\big[Y \mid D=0, S=1\big].$$

(3) **Upper bound (UB).** Trim away the *bottom* $(1-\alpha)$ of treated outcomes (i.e. assume the extra treated-only observed units are the *worst* outcomes). Then

$$\text{UB} \;=\; E\big[Y \mid D=1, S=1,\; Y \ge q_1(1-\alpha)\big] \;-\; E\big[Y \mid D=0, S=1\big].$$

```r
p1_hat <- mean(dat$S[dat$D==1])
p0_hat <- mean(dat$S[dat$D==0])

alpha <- p0_hat / p1_hat
cat("Retention fraction alpha =", round(alpha,4), "\n")
```

```
## Retention fraction alpha = 0.7536
```

```r
q_alpha   <- quantile(Y_treat_obs, probs = alpha)
q_1_alpha <- quantile(Y_treat_obs, probs = 1 - alpha)

treated_keep_low  <- Y_treat_obs[Y_treat_obs <= q_alpha]
treated_keep_high <- Y_treat_obs[Y_treat_obs >= q_1_alpha]

LB <- mean(treated_keep_low) - mean(Y_ctrl_obs)
UB <- mean(treated_keep_high) - mean(Y_ctrl_obs)
```

```r
cat("Lee Lower Bound:", round(LB,3), "\n")
```

```
## Lee Lower Bound: 0.342
```

```r
cat("Lee Upper Bound:", round(UB,3), "\n")
```

```
## Lee Upper Bound: 0.993
```

```r
cat("Naive observed difference:", round(naive_diff,3), "\n")
```

```
## Naive observed difference: 0.641
```

Simple bootstrap

```r
cat("95% CI for LB:", round(ci_LB,3), "\n")
```

```
## 95% CI for LB: 0.293
```

```r
cat("95% CI for UB:", round(ci_UB,3), "\n")
```

```
## 95% CI for UB: 1.04
```

Same set up but with $p_0 = 0.75, p_1 = 0.8$

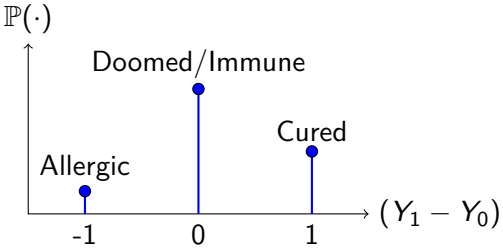
```
## Retention fraction alpha = 0.9468
## Lee Lower Bound: 0.341
## Lee Upper Bound: 0.512
## Naive observed difference: 0.641
## 95% CI for LB: 0.292
## 95% CI for UB: 0.574
```

Bounding the Distribution of Treatment Effects

- Randomly assign $D \implies$ ATE point identified: no selection bias!
- (Y_0, Y_1) never observed for same person; can't learn joint distribution.
- Anything that depends on this joint distribution is *not point identified*.
- Examples: $\text{Var}(Y_1 - Y_0)$, $\mathbb{P}(Y_1 - Y_0 > 0)$
- Can we *partially identify* the distribution of treatment effect $(Y_1 - Y_0)$?
- Start with binary Y case; then consider the general case.

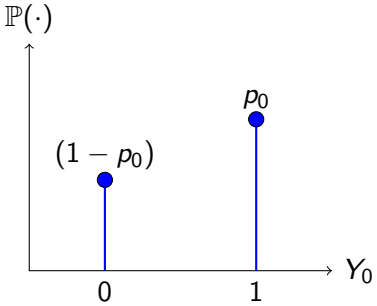
Unobserved: Joint Distribution of (Y_0, Y_1), Distribution of $(Y_1 - Y_0)$

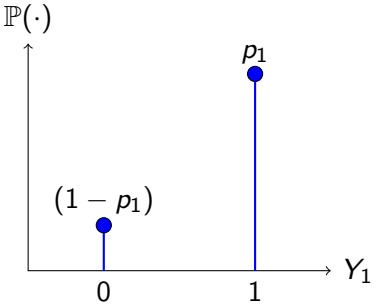
		Y_1	
		0	1
Y_0	0	$\mathbb{P}$(Doomed)	$\mathbb{P}$(Cured)
	1	$\mathbb{P}$(Allergic)	$\mathbb{P}$(Immune)



- Dangerous disease, and dangerous treatment.
- Treatment helps some people (the "Cured"), harms others (the "Allergic").
- Treatment has no effect on other people (the "Doomed" and "Immune")
- **Are more people helped than harmed?**

Observed: Marginal Distributions of Y_0 and Y_1





- Assume (Y, D) come from a randomized, double-blind, placebo-controlled trial.
- p_0 is the share of untreated who recover; p_1 is the share of treated who recover.
- The ATE is $p_1 - p_0$
- Try to bound what we can't observe using what we can observe.

From Joint (Unobserved) to Marginals (Observed)

Recall: $p_0 \equiv \mathbb{P}(Y_0 = 1)$ and $p_1 \equiv \mathbb{P}(Y_1 = 1)$.

		Y_1		
		0	1	
Y_0	0	$\mathbb{P}(\text{Doomed})$	$\mathbb{P}(\text{Cured})$	$\Longrightarrow \quad (1 - p_0) = \mathbb{P}(\text{D}) + \mathbb{P}(\text{C})$
	1	$\mathbb{P}(\text{Allergic})$	$\mathbb{P}(\text{Immune})$	$\Longrightarrow \quad p_0 = \mathbb{P}(\text{A}) + \mathbb{P}(\text{I})$

$$\Downarrow \qquad\qquad \Downarrow$$

$$(1 - p_1) = \mathbb{P}(\text{D}) + \mathbb{P}(\text{A}) \qquad p_1 = \mathbb{P}(\text{C}) + \mathbb{P}(\text{I})$$

Shorthand: $\alpha \equiv \mathbb{P}(\text{Allergic})$

Previous Slide

$$
\begin{aligned}
(1 - p_0) &= \mathbb{P}(\text{Doomed}) + \mathbb{P}(\text{Cured}) \\
p_0 &= \mathbb{P}(\text{Allergic}) + \mathbb{P}(\text{Immune}) \\
(1 - p_1) &= \mathbb{P}(\text{Doomed}) + \mathbb{P}(\text{Allergic}) \\
p_1 &= \mathbb{P}(\text{Cured}) + \mathbb{P}(\text{Immune})
\end{aligned}
$$

Rearranging

$$
\begin{aligned}
\mathbb{P}(\text{Immune}) &= p_0 - \alpha \\
\mathbb{P}(\text{Doomed}) &= (1 - p_1) - \alpha \\
\mathbb{P}(\text{Cured}) &= (p_1 - p_0) + \alpha
\end{aligned}
$$

- Everything is written in terms of observables (p_0, p_1) and α!

Bounding $\alpha \equiv \mathbb{P}(\text{Allergic})$

Previous Slide

- $\mathbb{P}(\text{Immune}) = p_0 - \alpha,\ \mathbb{P}(\text{Doomed}) = (1 - p_1) - \alpha,\ \mathbb{P}(\text{Cured}) = (p_1 - p_0) + \alpha$

Probabilities are between 0 and 1

- Apply Immune, Doomed, and Cured to bound α:

$$0 \leq (p_1 - p_0) + \alpha \leq 1, \quad 0 \leq (1 - p_1) - \alpha \leq 1, \quad 0 \leq p_0 - \alpha \leq 1.$$

Simplify

- Rearrange the preceding, and combine with $0 \leq \alpha \leq 1$

$$\max\{-\text{ATE}, 0\} \leq \alpha \leq \min\{p_0, (1 - p_1)\}, \quad \text{ATE} = (p_1 - p_0).$$

(Pointwise) Sharp Bounds for Distribution of Treatment Effects

Previous Slide

- $\mathbb{P}(\text{Immune}) = p_0 - \alpha,\ \mathbb{P}(\text{Doomed}) = (1 - p_1) - \alpha,\ \mathbb{P}(\text{Cured}) = (p_1 - p_0) + \alpha$
- $\max\{-(p_1 - p_0), 0\} \leq \alpha \leq \{p_0, (1 - p_1)\}$

Shorthand

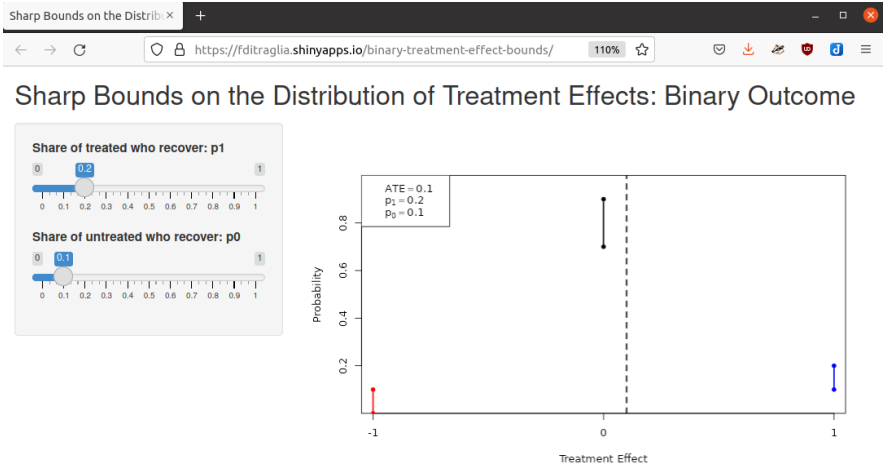
- $\underline{\alpha} \equiv \max\{-(p_1 - p_0), 0\}, \quad \overline{\alpha} \equiv \min\{p_0, (1 - p_1)\}$

Combine

- Recall that $\alpha \equiv \mathbb{P}(\text{Allergic}) = \mathbb{P}(Y_1 - Y_0 = -1)$

$$\underline{\alpha} \leq \mathbb{P}(Y_1 - Y_0 = -1) \leq \overline{\alpha}$$
$$(1 - p_1) + p_0 - 2\overline{\alpha} \leq \mathbb{P}(Y_1 - Y_0 = 0) \leq (1 - p_1) + p_0 - 2\underline{\alpha}$$
$$(p_1 - p_0) + \underline{\alpha} \leq \mathbb{P}(Y_1 - Y_0 = 1) \leq (p_1 - p_0) + \overline{\alpha}$$

https://fditraglia.shinyapps.io/binary-treatment-effect-bounds/



The General Case: Fan & Park (2010)

- Above we assumed that (Y_0, Y_1) were both binary.
- We asked which joint distributions were **not ruled out** based on the marginals.
- Pointwise sharp bounds for $\mathbb{P}(Y_1 - Y_0 = -1)$, $\mathbb{P}(Y_1 - Y_0 = 0)$ and $\mathbb{P}(Y_1 - Y_0 = 1)$.
- Special case of a general result: Fan and Park (2010).
- Same basic idea, but math is harder when (Y_0, Y_1) may not be binary.
- **This is a result you may actually use in practice!**
- Explain their result without proving it.

Fan & Park (2010) Bounds

Observables

- $F_0(y) \equiv \mathbb{P}(Y_0 \leq y)$ and $F_1(y) \equiv \mathbb{P}(Y_1 \leq y)$

Goal

- Sharp bounds for $F(\delta) \equiv \mathbb{P}(Y_1 - Y_0 \leq \delta)$

Notation

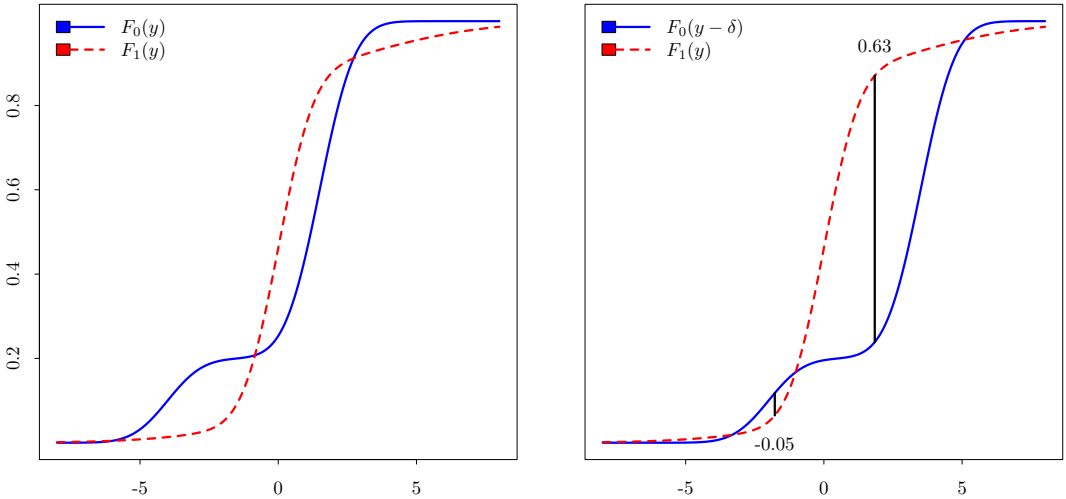
$$\underline{F}(\delta) \equiv \sup_y F_1(y) - F_0(y - \delta)$$

$$\overline{F}(\delta) \equiv 1 + \left[\inf_y F_1(y) - F_0(y - \delta)\right]$$

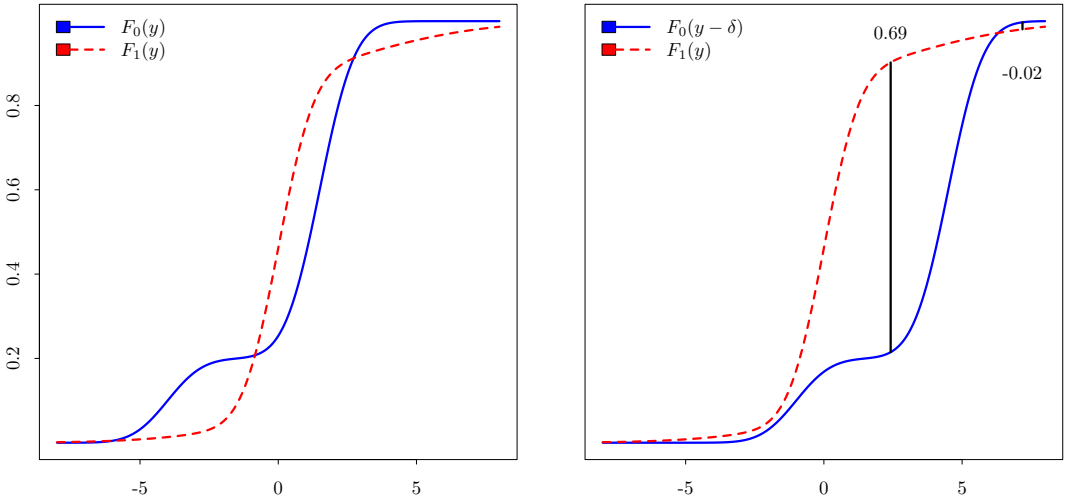
Theorem

- For any δ, $0 \leq \underline{F}(\delta) \leq F(\delta) \leq \overline{F}(\delta) \leq 1$. These bounds are (pointwise) sharp.

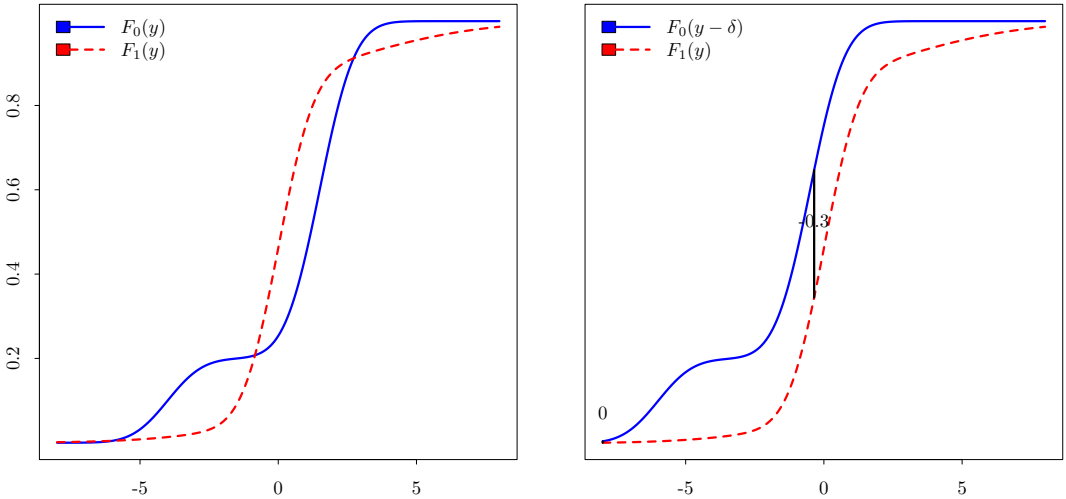
Left: $\delta = 0$, Right: $\delta = 2$



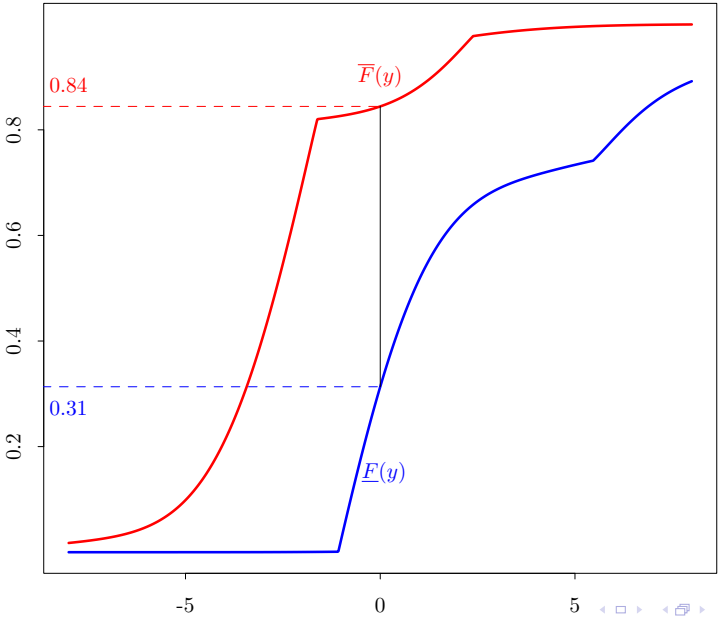
Left: $\delta = 0$, Right: $\delta = 3$



Left: $\delta = 0$, Right: $\delta = -2$



All the bounds!



Mogstad & Torgovitsky (2018)

Identification and Extrapolation of Causal Effects with Instrumental Variables, Mogstad & Torgovitsky (Annual Review of Economics, 2018)

- Instrumental variables (IV) address selection on unobservables by using exogenous variation.
- Standard IV identifies causal effects for those whose treatment is shifted by the instrument ("compliers").
- Policy questions often target different groups $\Rightarrow$ need to **extrapolate** beyond compliers.

Generalized Roy Model

Potential outcome framework with an instrument:

$$Z(Y_1, Y_0, D_1, D_0)|X$$

Imbens–Angrist monotonicity with a weakly separable choice model:

$$D = 1[\nu(X, Z) - U \geq 0] \tag{2}$$

Normalize $U|X \sim \mathrm{Unif}[0, 1]$ so that the propensity score is $p(X, Z)$:

$$D = 1[U \leq p(X, Z)], \quad U \sim \mathrm{Unif}[0, 1] \tag{4}$$

MTR and MTE

Marginal Treatment Response (MTR):

$$m_0(u,x) = \mathbb{E}[Y_0 \mid U = u, X = x], \quad m_1(u,x) = \mathbb{E}[Y_1 \mid U = u, X = x] \tag{6}$$

Marginal Treatment Effect (MTE):

$$\mathrm{MTE}(u,x) = \mathbb{E}[Y_1 - Y_0 \mid U = u, X = x] \tag{5}$$

Captures (observed and unobserved) heterogeneity in treatment effects over U and X.

Target Parameters as Weighted Averages of MTRs

Let $\omega_0^*(u, X, Z)$ and $\omega_1^*(u, X, Z)$ be identified weights. Define:

$$\beta^* = \mathbb{E}\left[\int_0^1 m_0(u, X)\, \omega_0^*(u, X, Z)\, du\right] + \mathbb{E}\left[\int_0^1 m_1(u, X)\, \omega_1^*(u, X, Z)\, du\right] \tag{8}$$

Different choices of weights recover ATE, ATT, ATU, LATEs, and PRTEs.

Table 1: Weights for Conventional Treatment Parameters

Target parameter	**Expression**	$\omega_0^*(\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{z})$	$\omega_1^*(\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{z})$
Average untreated outcome	$\mathbb{E}[Y_0]$	1	0
Average treated outcome	$\mathbb{E}[Y_1]$	0	1
ATE	$\mathbb{E}[Y_1 - Y_0]$	-1	1
ATE given $X = \bar{x}$, $P[X = \bar{x}] > 0$	$\mathbb{E}[Y_1 - Y_0 \mid X = \bar{x}]$	$-\omega_1^*(u, x, z)$	$\frac{\mathbf{1}[x = \bar{x}]}{P[X = \bar{x}]}$
ATT	$\mathbb{E}[Y_1 - Y_0 \mid D = 1]$	$-\omega_1^*(u, x, z)$	$\frac{\mathbf{1}[u \leq p(x, z)]}{P[D = 1]}$
ATU	$\mathbb{E}[Y_1 - Y_0 \mid D = 0]$	$-\omega_1^*(u, x, z)$	$\frac{\mathbf{1}[u > p(x, z)]}{P[D = 0]}$
LATE for $z_0 \to z_1$ given $X = x$	$\mathbb{E}[Y_1 - Y_0 \mid p(x, z_0) < U \leq p(x, z_1),\ X = x]$	$-\omega_1^*(u, x, z)$	$\frac{\mathbf{1}[\, p(x, z_0) < u \leq p(x, z_1)\,]}{p(x, z_1) - p(x, z_0)}$

Table: Weights for conventional treatment effect parameters.

Policy-Relevant Treatment Effects (PRTE)

Compare two policies a_1 and a_0 that change participation:

$$\mathrm{PRTE} = \frac{\mathbb{E}[Y^{a_1}] - \mathbb{E}[Y^{a_0}]}{\mathbb{E}[D^{a_1}] - \mathbb{E}[D^{a_0}]} \tag{12}$$

LATE is a special case: assign everyone $Z = z_1$ vs. $Z = z_0$:

$$\mathbb{E}[Y_1 - Y_0 \mid p(z_0) < U \leq p(z_1)] \tag{13}$$

Table 2: Weights for PRTEs

Target parameter	Expression	Weights
Generalized LATE for $U \in (\underline{u}, \overline{u}]$	$\mathbb{E}[Y_1 - Y_0 \mid U \in (\underline{u}, \overline{u}]]$	$\omega_1^*(u,x,z) = \dfrac{\mathbf{1}[u \in (\underline{u}, \overline{u}]]}{\overline{u} - \underline{u}}, \quad \omega_0^* = -\omega_1^*$
PRTE for policy (p^{a_1}, Z^{a_1}) relative to (p^{a_0}, Z^{a_0})	$\dfrac{\mathbb{E}[Y^{a_1}] - \mathbb{E}[Y^{a_0}]}{\mathbb{E}[D^{a_1}] - \mathbb{E}[D^{a_0}]}$	$\omega_1^*(u,x,z) = \dfrac{P[u \leq p^{a_1}(x, Z^{a_1}) \mid X = x] - P[u \leq p^{a_0}(x, Z^{a_0}) \mid X = x]}{\mathbb{E}[p^{a_1}(X, Z^{a_1})] - \mathbb{E}[p^{a_0}(X, Z^{a_0})]}, \ \omega_0^* = -\omega_1^*$
Additive PRTE with magnitude α ($Z^a = Z$, $p^a = p + \alpha$)	PRTE as above	$\omega_1^*(u,x,z) = \dfrac{\mathbf{1}[u \leq p(x,z) + \alpha] - \mathbf{1}[u \leq p(x,z)]}{\alpha}, \ \omega_0^* = -\omega_1^*$
Proportional PRTE with magnitude α ($p^a = (1+\alpha)p$)	PRTE as above	$\omega_1^*(u,x,z) = \dfrac{\mathbf{1}[u \leq (1+\alpha)p(x,z)] - \mathbf{1}[u \leq p(x,z)]}{\alpha\, \mathbb{E}[p(X,Z)]}, \ \omega_0^* = -\omega_1^*$
PRTE for additive α shift of jth component of Z	$Z^a = Z + \alpha e_j$	$\omega_1^*(u,x,z) = \dfrac{\mathbf{1}[u \leq p(x, z + \alpha e_j)] - \mathbf{1}[u \leq p(x,z)]}{\mathbb{E}[p(X, Z + \alpha e_j)] - \mathbb{E}[p(X,Z)]}, \ \omega_0^* = -\omega_1^*$

Table: Weights for PRTEs.

Nonparametric Point Identification: Two Special Cases

We know of two special cases where we can identify the parameters of interest:

- LATEs: Wald estimand identifies $E[Y_1 - Y_0 \mid p(x, z_0) < U \leq p(x, z_1), X = x]$.
- Continuous $P = p(X, Z)$ with support covering weights: Local IV identifies MTE at interior points.

IV-like Estimands (What We Know)

Example: linear IV slope

$$\beta_{IV} = \frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(D, Z)} \tag{18}$$

More generally, for a measurable specification $s(d, x, z)$:

$$\beta_s = \mathbb{E}[\, s(D, X, Z)\, Y\,] \tag{20}$$

Each β_s can be written as a weighted average of the MTRs.

Common IV-like Estimands

Estimand β_s	**Definition**	$s(d, x, z)$ **(IV-like specification)**
Wald ($z_0 \to z_1$)	$\dfrac{\mathbb{E}[Y \mid Z = z_1] - \mathbb{E}[Y \mid Z = z_0]}{\mathbb{E}[D \mid Z = z_1] - \mathbb{E}[D \mid Z = z_0]}$	$\dfrac{\mathbf{1}[z = z_1]/P[Z = z_1] - \mathbf{1}[z = z_0]/P[Z = z_0]}{\mathbb{E}[D \mid Z = z_1] - \mathbb{E}[D \mid Z = z_0]}$
IV slope (scalar Z)	$\dfrac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(D, Z)}$	$\dfrac{z - \mathbb{E}[Z]}{\mathrm{Cov}(D, Z)}$
OLS slope for D	$\dfrac{\mathrm{Cov}(Y, D)}{\mathrm{Var}(D)}$	$\dfrac{d - \mathbb{E}[D]}{\mathrm{Var}(D)}$
OLS, jth component of $X^\sim$	$e_j^\top \left(\mathbb{E}[X^\sim X^{\sim\top}]\right)^{-1} \mathbb{E}[X^\sim Y]$	$e_j^\top \left(\mathbb{E}[X^\sim X^{\sim\top}]\right)^{-1} x^\sim$
TSLS, jth component	$e_j^\top \left(\mathbb{E}[Z^\sim X^{\sim\top}]\right)^{-1} \mathbb{E}[Z^\sim Y]$	$e_j^\top \left(\mathbb{E}[Z^\sim X^{\sim\top}]\right)^{-1} z^\sim$

$X^\sim = [1,\ D,\ X^\top]^\top, \quad Z^\sim = [1,\ Z,\ X^\top]^\top, \quad e_j$ is the jth unit vector.

Table: Common IV-like estimands and their $s(d, x, z)$ specifications.

Weight Representation of IV-like Estimands

For any s,

$$\beta_s = \mathbb{E}\left[\int_0^1 m_0(u, X)\, \omega_0^s(u, X, Z)\, du + \int_0^1 m_1(u, X)\, \omega_1^s(u, X, Z)\, du\right] \tag{20}$$

with

$$\omega_0^s(u, x, z) = s(0, x, z)\, \mathbf{1}[u > p(x, z)], \quad \omega_1^s(u, x, z) = s(1, x, z)\, \mathbf{1}[u \leq p(x, z)].$$

From What We Know to What We Want

Let M be an admissible set of MTR pairs $m = (m_0, m_1)$.

$$M_S = \{\, m \in M : \; s(m) = \beta_s \;\; \forall s \in S \,\} \tag{24}$$

Induced set of feasible target parameters:

$$B_S^* = \{\, b \in \mathbb{R} : \; b = \beta^*(m) \text{ for some } m \in M_S \,\} \tag{25}$$

Bounds via optimization:

$$\underline{\beta}^* = \inf_{m \in M} \beta^*(m) \quad \text{s.t. } s(m) = \beta_s \;\; \forall s \in S \tag{1}$$

$$\overline{\beta}^* = \sup_{m \in M} \beta^*(m) \quad \text{s.t. } s(m) = \beta_s \;\; \forall s \in S \tag{26}$$